Angeline Aguinaldo

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Research Areas

Embodied AI and robotics; task plan transfer (sequential, hierarchical, and analogical) across planning domains, abstraction levels, and robot embodiments; ontology-driven plan adaptation; structure-preserving methods for compositional task planning and reasoning; safety-critical autonomous systems; categorical foundations for AI and robotics.

Education

Ph.D., Computer Science, University of Maryland, College Park, 2025

Dissertation: *Sequential, Hierarchical, and Analogical Plan Transfer in Robotics*

Advisor: William Regli

M.S., Electrical Engineering, Drexel University, 2017

B.S., Biomedical Engineering, Drexel University, 2017

Technical Skills

Languages: Python, Julia, PDDL, SQL
ML & Deep Learning: PyTorch, TensorFlow, NumPy, SciPy, pandas, scikit-learn, OpenCV
Robotics & Simulation: ROS, Gazebo, AI2-THOR, MoveIt
Software Engineering: Git, Docker, RunAI, FastAPI, CrewAI, PydanticAI, LangGraph

Selected Awards & Honors

2025, **Larry S. Davis Doctoral Dissertation Award**, University of Maryland Department of Computer Science

2023, Best Paper Award, AAAI Fall Symposium on Unifying Representations for Robot Application Development (URRAD)

2018–2021, 2026, Special Achievement Award, Johns Hopkins University Applied Physics Laboratory

Experience

Ph.D., Computer Science, University of Maryland, College Park Sept 2018 – May 2025

- Developed a formal framework for sequential, hierarchical, and analogical plan transfer across planning domains, abstraction levels, and robot embodiments
- Developed category-theoretic representations (C-sets, double-pushout rewriting, functorial data migrations) for structured reasoning over action operators in symbolic planning domains
- Formalized hierarchical plan composition via functorial semantics; validated on Baxter and UR5 arms in a machine-tending scenario (Siemens collaboration)
- Demonstrated cross-domain plan transfer via ontology-guided alignment (Blocksworld → AI2-THOR)

Senior Research Scientist, JHU Applied Physics Laboratory Aug 2017 – Present

- Led research on collaborative agent architectures for long-horizon planning and decision support

- Built a simulation environment for sequential planning under uncertainty with probabilistic dynamics and batch policy evaluation
- Developed formal behavior representations for safety-critical autonomous systems; co-authored a NASA Technical Memorandum on category-theoretic systems ontology for aviation autonomy *[link]*
- Led CV pipelines and scalable ML infrastructure for daily disaster-response products to FEMA; flood segmentation across 2,000+ km^2 of multispectral imagery (2020 Fast Company World Changing Ideas Finalist) *[link]*
- People manager, Human-AI Collaboration group (8 researchers and engineers); hiring, career development, and technical mentorship

Research Associate, Topos Institute 2022 – 2023

- Developed categorical foundations for plan transfer; co-authored the Topos *Relational Thinking* textbook

Computer Vision Researcher, Bioimage Lab, Drexel University 2015 – 2017

- Developed elliptical shape models for cell separation in live-cell microscopy; published in IEEE Transactions on Medical Imaging

Photonics Summer REU, Princeton University 2013

Publications

Peer-Reviewed Articles

- **Aguinaldo, A.**, Patterson, E., Regli, W. Automating Transfer of Robot Task Plans Using Functorial Data Migrations. *IEEE Transactions on Automation Science and Engineering*, 2025 *[paper]*
- **Aguinaldo, A.**, Patterson, E., Fairbanks, J., Regli, W., Ruiz, J. A Categorical Representation Language and Computational System for Knowledge-Based Planning. *AAAI Fall Symposium on Unifying Representations for Robot Application Development (URRAD)*, 2023. **Best Paper** *[paper]*
- **Aguinaldo, A.**, Bunker, J., Pollard, B., Shukla, A., Canedo, A., Quiros, G., Regli, W. RoboCat: A Category Theoretic Framework for Robotic Interoperability Using Goal-Oriented Programming. *IEEE Transactions on Automation Science and Engineering*, 2021 *[paper]*
- Winter, M., Mankowski, W., Wait, E., De La Hoz, E. C., **Aguinaldo, A.**, Cohen, A. R. Separating Touching Cells Using Pixel Replicated Elliptical Shape Models. *IEEE Transactions on Medical Imaging*, 38(4):883–893, 2019
- Player, R. A., **Aguinaldo, A.**, Merritt, B. B., Maszkiewicz, L. N., Adeyemo, O. E., Forsyth, E. R., Verratti, K. J., Chee, B. W., Grady, S. L., Bradburne, C. E. The META Tool Optimizes Metagenomic Analyses Across Sequencing Platforms and Classifiers. *Frontiers in Bioinformatics*, 2:969247, 2023 *[paper]*
- Caino, M., Seo, J., **Aguinaldo, A.**, et al. A neuronal network of mitochondrial dynamics regulates metastasis. *Nature Communications*, 7:13730, 2016

Technical Reports

- Mahlum, M., Jarvis, S., Niu, N., **Aguinaldo, A.**, Hicks, A., Levitt, I. Formal Structures in Systems Ontology towards Air Traffic Management Architectures. NASA Technical Memorandum NASA/TM-20250010771, November 2025 *[paper]*

Workshop Papers & Preprints

- **Aguinaldo, A.** Functorial Plan Transfer: Compositional Generalization for LLM Agents via OpenAPI Schema Categories. *In preparation*
- **Aguinaldo, A.**, Regli, W. Encoding Compositionality in Classical Planning Solutions. *IJCAI Workshop on Generalization in Planning*, 2021 *[paper]*
- **Aguinaldo, A.**, Regli, W. A Graphical Model-Based Representation for Classical AI Plans using Category Theory. *ICAPS Workshop on Explainable AI Planning*, 2021
- **Aguinaldo, A.**, Chiang, P., Gain, A., Patil, A., Pearson, K., Feizi, S. Compressing GANs using Knowledge Distillation. arXiv preprint, 2019 *[paper]*

Books

- Co-author, *Relational Thinking: From Abstractions to Applications.* Digital textbook, Topos Institute, 2024 *[book]*

Selected Talks & Presentations

- AMS Joint Mathematics Meetings, Applied Category Theory Special Session, "Analogical Plan Transfer in Robotics using Functorial Data Migrations", 2025 *[slides]*
- JuliaCon, "What's the plan?", asked my robot, 2024 *[video]*
- Microsoft Future Leaders in Robotics and AI Seminar Series, "A Category Theoretic Approach to Planning in a Complex World", 2023 *[video]*
- Topos Institute Berkeley Seminar, "Category theory for automated planning and program compilation in robotics", 2022 *[video]*
- NIST Compositional Structures for Systems Engineering and Design Workshop, "Contextual affordances in context-aware autonomous systems", 2022
- ICRA Compositional Robotics Workshop, "Modeling traceability, change information, and synthesis in autonomous system design using symmetric delta lenses", 2022
- Xerox PARC Design Seminar, "Applications of category theory to automated planning and program compilation in robotics", 2022
- IJCAI Workshop on Generalization in Planning, "Encoding Compositionality in Classical Planning Solutions", 2021
- ICAPS Workshop on Explainable AI Planning, "A Graphical Model-Based Representation for Classical AI Plans using Category Theory", 2021

Professional Service & Community Leadership

- Co-organizer, Applied Category Theory Special Session, AMS Joint Mathematics Meetings, 2026
- Seminar Series Chair, Agentic AI and Human-AI Collaboration, JHU Applied Physics Laboratory, 2024–2026
- Co-organizer, The Adjoint School, 2022, 2023
- Local Organizer, Applied Category Theory (ACT) Conference, 2023
- Co-organizer, ICRA Compositional Robotics: Mathematics and Tools Workshop, 2023
- Student, The Adjoint School, 2021
- Reviewer, International Conference on Automated Planning and Scheduling (ICAPS)
- Reviewer, *Mathematical Structures in Computer Science*
- Open-source contributor, AlgebraicJulia